



Unidad Profesional
Interdisciplinaria en Ingeniería y
Tecnologías Avanzadas

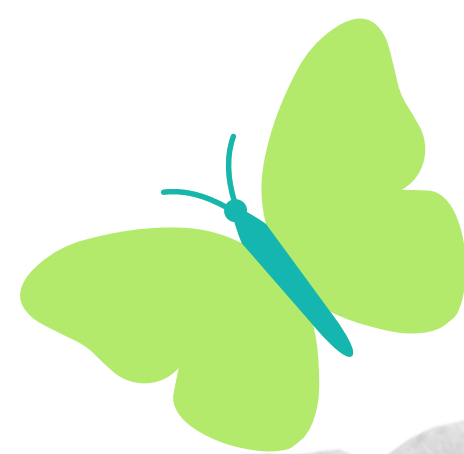
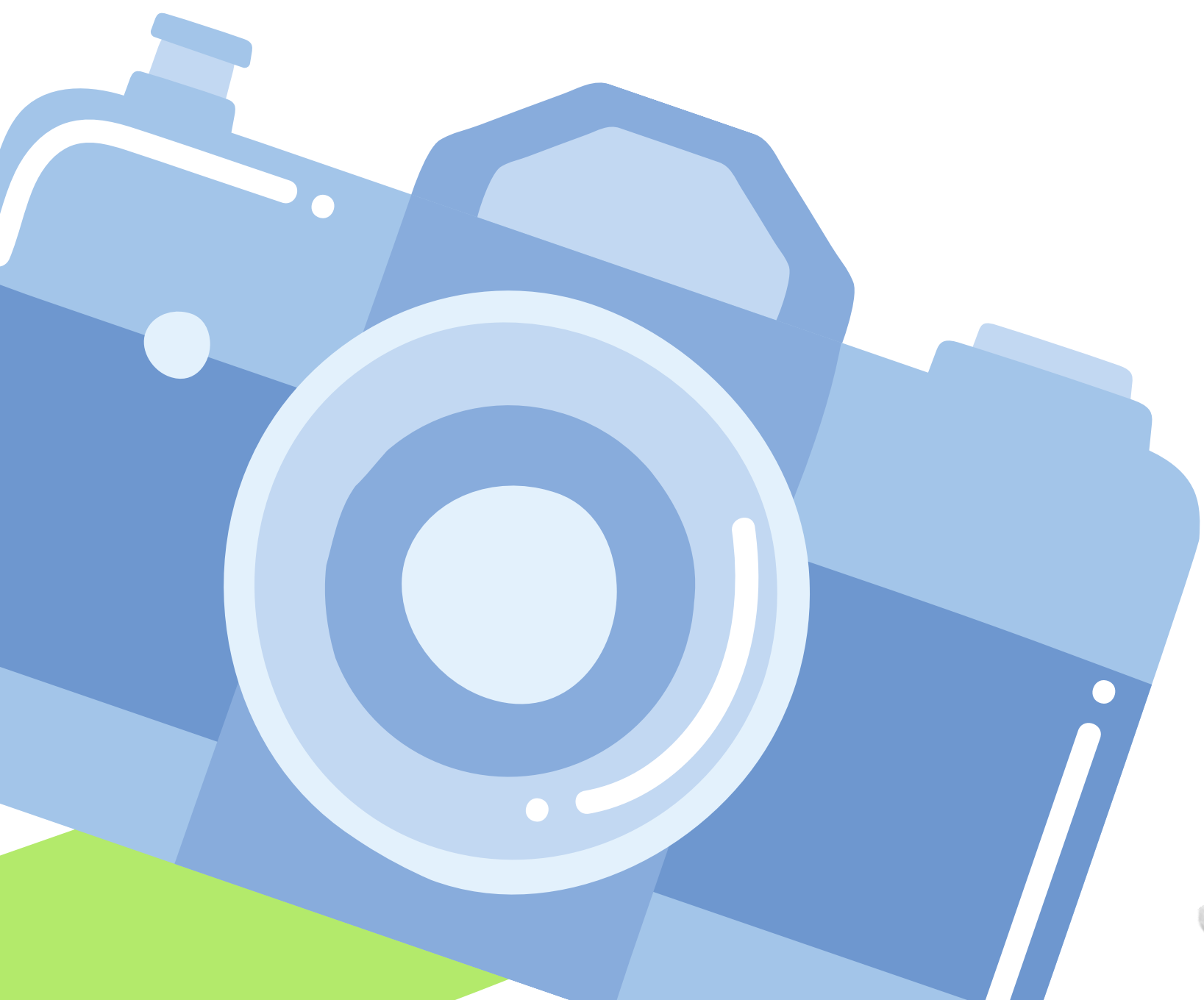
upiiita

Multimedia

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Práctica 15: cambio de tamaño



Código

```
import struct
import random

def random_pixel_bytes():
    """Genera 3 bytes aleatorios para un píxel (BGR)."""
    return random.randbytes(3)

def expand_bmp_preserve_content(input_path, output_path):
    with open(input_path, "rb") as f:
        header = bytearray(f.read(54))

    orig_w = struct.unpack('<l', header[18:22])[0]
    orig_h = struct.unpack('<l', header[22:26])[0]

    f.seek(54)
    orig_pixel_bytes = f.read(orig_w * orig_h * 3)

    new_w = orig_w + 1
    new_h = orig_h + 1

    row_bytes_data = new_w * 3
    padding_bytes = (4 - (row_bytes_data % 4)) % 4

    new_pixel_data = bytearray()

    for row in range(orig_h):
        start = row * orig_w * 3
        end = start + (orig_w * 3)
        row_pixels = orig_pixel_bytes[start:end]

        new_pixel_data.extend(row_pixels)
        new_pixel_data.extend(random_pixel_bytes())

        new_pixel_data.extend(b'\x00' * padding_bytes)

    for _ in range(new_w):
        new_pixel_data.extend(random_pixel_bytes())
        new_pixel_data.extend(b'\x00' * padding_bytes)

    new_file_size = 54 + len(new_pixel_data)
    header[2:6] = struct.pack('<l', new_file_size)
    header[18:22] = struct.pack('<l', new_w)
    header[22:26] = struct.pack('<l', new_h)

    with open(output_path, "wb") as f:
        f.write(header)
        f.write(new_pixel_data)

    print(f"Imagen guardada con éxito en '{output_path}'.")
    print(f"Dimensiones: {new_w}x{new_h} píxeles.")
    print(f"Nuevo tamaño total del archivo: {new_file_size} bytes.")

expand_bmp_preserve_content("./images/example001.bmp",
"./images/expanded_17x17píxeles.bmp")
```

Resultado

